
HIMANI CHOUDHARY

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Experienced bioinformatician, with a proven track record of transforming complex biological datasets into actionable insights, developing automated analysis pipelines, and driving research innovation through advanced statistical modeling. Skilled in Python, R, Perl and Bash programming, machine learning, deep learning and data visualization. Looking for a new opportunity to apply my skills in the field of data science.

SKILLS

PROGRAMMING:

Advance Python (OOPs), R Programming, MATLAB, Perl, Unix, Linux, Bash, PHP, HTML, MySQL(basic), Nextflow, Docker and Git.

MACHINE LEARNING, DEEP LEARNING AND DATA VISUALIZATION:

NumPy, Pandas, Sklearn, Matplotlib, ggplot2, seaborn, grid, lattice, kNN, DT, SVM, RF, Lasso regression, Adaboost, RFECV, Logistic regression, ExtraTrees, Scikit, Tableau, Tensorflow, Keras, CNN.

WEB FRAMEWORK: FastAPI

GENOMICS, TRANSCRIPTOMICS, METAGENOMICS, EPIGENETICS AND MUTI-OMICS:

WGS, RNA-Seq, ChIP-Seq, ATAC-seq, amplicon seq, shotgun seq
Tools : Velvet, BWA, Bowtie2, SAMtools, VCFtools, GATK, Trinity, MEGAHIT, SPAdes, HISAT2, STAR, DEGseq2, DADA2, QIIME2, and MACS2.

STRUCTURAL BIOINFORMATICS:

VAST, DALI, BLAST, PSI-BLAST, PDB, PDBsum, AlphaFold, DiscoveryStudio, PyMol.

DATABASES AND SERVERS:

NCBI, Ensembl, JBrowse, UCSC Genome Browser, TCGA, SRA, GEO, Silva BLAST.

Excellent communication skills (vocal and written) coupled with great time-management skills.

AWARDS:

GAT-B: Qualified (AIR-72)
DBT Scholarship: 2022 – 2024
GATE XL (2024) - Qualified

EXPERIENCE

Edgene Biomed Pvt Ltd | Technical Head - Bioinformatics

Mar 2024 – Present | Gurugram

- Designed and deployed a **transcriptomic data analysis report generation platform** using Python, with fully functional APIs for seamless integration.
- Built and deployed a **metagenomic data analysis report generation platform** using R, enabling automated generation of high-quality PDF reports.
- Led collaboration and mentorship efforts for developing an **automated metagenomic analysis pipeline** using **QIIME2** and **Bash scripting**.
- Developed a **machine learning-based biomarker prediction model utilizing gene expression data** for diagnostic and prognostic insights.

PROJECTS

INTERNATIONAL PROJECT

Blended Mobility Focusing on Norway-India Water-Soil Microbiome Nexus (NIWASm) in collaboration with University of Southeastern Norway.

Bioinformatics Centre, SPPU; University of Southeastern Norway, BO, Telemark, Norway; (Mar 2023 – Present)

MASTER THESIS

Time series approach to detect structural signatures in proteins using conformational states potentials

Bioinformatics Centre, SPPU; Department of Statistics, SPPU, Pune; (Nov 2023 – May 2024)

MACHINE LEARNING AND TRANSCRIPTOMICS

Predictive modeling for breast cancer classification using gene expression profiles and machine learning techniques

DEEP LEARNING

Potato plant disease detection using cnn-based image classification

EDUCATION

M.SC. BIOINFORMATICS

Bioinformatics Centre, Savitribai Phule Pune University, Pune
CGPA – 9.6

Aug 2022 – May 2024

B.SC. BIOTECHNOLOGY

Department Of Biotechnology, Central University of Rajasthan
CGPA – 8.15

July 2019 – May 2022

CERTIFICATIONS

WORKSHOP ON “GENERATIVE AI APPLICATIONS IN SCIENTIFIC RESEARCH AND INDUSTRIAL PROBLEMS”:

The workshop provided an introduction to fundamental AI/ML/DL techniques and their practical implementation in the development of generative AI servers. It also explored their relevance in biological research, particularly in precision medicine and the use of generative AI for disease diagnosis.

BIOINFORMATICS CENTRE, SPPU; DEPARTMENT OF MATHEMATICS, SPPU; PERSISTENT SYSTEMS, PUNE (JAN 2024)

WORKSHOP AND SYMPOSIUM AS A PART OF UTFORSK PROJECT, “BLENDED MOBILITY FOCUSING ON NORWAY INDIA WATER SOIL MICROBIOME”:

The workshop featured a series of lectures, project discussions, and interactive sessions focusing on the exploration of scientific research concepts related to the soil-water microbiome nexus between Norway and India.

DEPARTMENT OF NATURAL SCIENCES, UNIVERSITY OF SOUTHEASTERN NORWAY, BO, TELEMARK, NORWAY (SEPT 2023)

WORKSHOP AND SYMPOSIUM AS A PART OF UTFORSK PROJECT, “BLENDED MOBILITY FOCUSING ON NORWAY INDIA WATER SOIL MICROBIOME”:

The workshop covered the basics of metagenomic data analysis, including various pipelines used to extract different analytical insights from the data. This included an overview of the DADA2 pipeline, which is based on the R programming language. Additionally, we were introduced to geological survey data related to soil, and there were also opportunities for site visits.

BIOINFORMATICS CENTRE, SPPU (MARCH 2023)

HANDS-ON WORKSHOP ON “NGS DATA ANALYSIS OF VIRAL AND CLINICAL DATA”:

Sessions on Illumina informatics solutions, hands-on COVID data analysis, as well as demonstrations of exome data analysis and variant interpretation techniques.

PREMAS LIFE SCIENCES, DELHI (MARCH 2023)

ONLINE COURSE ON “ENGINEERING MATHEMATICS”:

The course covered various topics in differential calculus, integral calculus, linear algebra and differential equations with applications to various engineering problems.

IIT KHARAGPUR (NPTEL)

DEC – 2021 (12 WEEKS)

ONLINE COURSE ON “COMPUTER AIDED DRUG DESIGN”:

The course covered structure and target-based design, molecular modeling, quantum mechanics, drug likeness properties, QSAR and pharmacokinetic and dynamics using several software like Autodock, MarvinSketch.

IIT MADRAS (NPTEL)

DEC – 2020 (8 WEEKS)